



The business impact of an integrated continuous biomanufacturing platform for recombinant protein production



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ABSTRACT

The biotechnology industry primarily uses batch technologies to manufacture recombinant proteins. The natural evolution of other industries has shown that transitioning from batch to continuous processing can yield significant benefits. A quantitative understanding of these benefits is critical to guide the implementation of continuous processing. In this manuscript, we use process economic modeling and Monte Carlo simulations to evaluate an integrated continuous biomanufacturing (ICB) platform and conduct risk-based valuation to generate a probabilistic range of net-present values (NPVs). For a specific ten-year product portfolio, the ICB platform reduces average cost by 55% compared to conventional batch processing, considering both capital and operating expenses. The model predicts that these savings can further increase by an additional 25% in situations with higher-than-expected product demand showing the upward potential of the ICB platform. The ICB platform achieves these savings and corresponding flexibility mainly due to process intensification in both upstream and downstream unit operations. This study demonstrates the promise of continuous bioprocessing while also establishing a novel framework to quantify financial benefits of other platform process technologies.

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1. Introduction

The biotechnology industry is relatively young, beginning with the commercial launch of recombinant insulin and monoclonal antibodies in the 1980s. Over the next twenty years, the industry grew rapidly and focused on bringing innovative products to the market. This era of product innovation led to high revenues and large profit margins, resulting in the establishment of a manufacturing technology base with little regard for cost and effectiveness of manufacturing assets.

As the industry has matured, it has increasingly recognized that there are major issues with the structure and cost of these manufacturing approaches (Farid, 2007). Extensive research has improved understanding around the costs of goods (COGs) for recombinant protein production, leading to large reductions (as much as 100-fold) in operating expenses via process improvements and operational efficiencies (Sinclair and Monge, 2002; Rathore et al., 2004; Werner, 2004; Rajapakse et al., 2005; Farid, 2013). Key examples of process improvements include cell culture titer increases

(Croughan, 2008) and improved downstream yields (Gronemeyer et al., 2014). Examples of operational efficiencies include template platform processes (Kelley, 2007; Shukla and Thömmes, 2010) and operational improvement programs (Han et al., 2010) allowing better utilization of existing infrastructure. Collectively, this work has been a celebrated success for cost engineers, development scientists and operations groups in the industry.

However, biotechnology companies are now facing a new set of business realities and uncertainties that include adapting to potential competition after patent expiry, supplying complex and rapidly evolving biologics portfolios and driving growth through patient access beyond current mature markets (Gottschalk et al., 2013; Love et al., 2013; Ernst and Young, 2014). (For clarity, in this manuscript, we focus only on bioprocess development and specifically omit challenges in discovery and clinical research.) In the face of this changing landscape, two common needs for future biomanufacturing are emerging: increased flexibility and reduced cost of goods. Manufacturing flexibility allows companies to manage a complex and evolving portfolio where product numbers, volumes and types are always in flux due to scientific and market uncertainties, and mergers and acquisitions. Although operating expenses for some biologics products have decreased considerably, manufacturing facility construction requires significant time and capital, and operating expenses remain high for many non-standard products.

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Table 1
Hypothetical product launch scenario.

Product type	Launch date	Annual demand (kg/yr)
mAb	2025	200
non-mAb	2026	20
mAb	2027	200
non-mAb	2029	20
mAb	2029	200
mAb	2031	200
non-mAb	2031	20
mAb	2033	200
non-mAb	2034	20
mAb	2035	200

If additional manufacturing must be co-localized with new patient markets, simplified facility design and reduced capital investments become even more critical.

These challenges may appear unique to the biologics business but, in our view, are integral to the business lifecycle of many industries and offer similar opportunities to spur innovations in process development. Many industries successfully transition from batch processing to continuous processing to maximize flexibility and minimize cost of goods while still maintaining operational excellence (Tanner, 1998; Thomas, 2008; Reay et al., 2013). Other benefits that typically accompany this transition include standardization, simplified scale-up, and more consistent product quality (Anderson, 2001). Recently, several of these benefits have been qualitatively described and explored for the biotechnology industry (Baker, 2013; Weintraub, 2013; Whitford and Sargent, 2013; Konstantinov and Cooney, 2014). A quantitative understanding of these benefits is critical to drive process and technology development and organizational decision-making.

In this manuscript, we describe an integrated continuous biomanufacturing (ICB) platform for the production of drug substance with robust product quality and propose a novel methodology to quantify its benefits and develop a business case via comparison to conventional batch processing. Previous research has focused on continuous upstream (Pollock et al., 2012), continuous downstream (Pollock et al., 2013) and continuous processing for monoclonal antibodies (Biopharm Services, 2014). Because evaluation of individual unit operations or individual products can lead to biased technology selection that may result in a suboptimal biomanufacturing strategy, we holistically compare entire platforms (comprising all unit operations) together with a complex product portfolio. We also probe the relative flexibility of the ICB platform by evaluating the impact of several business and technical uncertainties, including product type, product approval, product demand and technology transfer delays. Overall, our work establishes a new way to build a business case for bioprocess platform technology selection and reveals the potential of continuous bioprocessing.

2. Methods

2.1. Product launch scenario

In this study, we assumed a hypothetical, mature product portfolio of ten products transitioning into phase III over ten years (Table 1). Because current protein therapeutics have varying degrees of stability and demand, we assumed two different product types: a more stable, high-demand product (such as a monoclonal antibody) and a less stable, low-demand product (such as enzymes, growth factors or fusion proteins). In the product launch scenario, we generically use the terms mAb and non-mAb to refer to these stable and less stable product types, respectively. Annual demands

of 200 kg for mAb products and 20 kg for non-mAb products were based on industry averages (Kelley, 2009; Aggarwal, 2014).

2.2. Platforms and processes

In this study, we evaluated a novel ICB platform that couples 500-L single-use reactors to a continuous capture operation. We chose a 500-L working volume because this volume is well positioned to serve both low- and high-demand products. For mAb production, the entire process is fully continuous from production bioreactor to drug substance, including post-capture purification and filtration. Because non-mAb purification cannot necessarily rely on affinity chromatography and typically has a more complicated downstream process architecture, we designed the non-mAb facility such that the continuous capture step is followed by batch intermediate and polishing purification and filtration. (Hereafter, we refer to this combination of continuous and batch operations as hybrid purification.)

To understand potential advantages and disadvantages of the continuous process, we compared it to a conventional, established bioprocessing platform. Since mAbs are typically manufactured in facilities built around large fed-batch bioreactors and large batch purification operations, (Kelley, 2009) the conventional platform uses 10,000-L stainless-steel reactors followed by large-diameter columns. However, for less stable products, batch bioreactors are not feasible, and other approaches such as perfusion cell culture are currently used (Pollock et al., 2012). There are multiple approaches to non-mAb manufacturing, but one example of a conventional non-mAb platform utilizes 2000-L stainless-steel perfusion bioreactors with microcarrier-adherent cell cultures followed by batch purification (Pattison et al., 2000). The two platforms and their processes are summarized in Table 2. The high level assumptions around each process are summarized in Table 3. These process assumptions correspond to capabilities of conventional technologies and the recently developed, novel ICB platform (Warikoo et al., 2012; Godawat et al., 2012). We performed mass balances for each of these processes to appropriately scale necessary equipment. Schematics of the unit operations in each process are shown in Fig. 1 and are described in detail in the Supplementary information (S1).

One of the advantages of single-use bioreactor technology is that facilities can be deployed in a modular fashion as needed (Levine et al., 2013). For the ICB mAb facility, we assumed that the first module consists of one reactor and the downstream train. As the product launch scenario progresses and increased throughput is required, two additional modules can be built, each consisting of another 500-L reactor. These multiple reactors can feed simultaneously into the same train (Fig. 2A). The 500-L non-mAb facility has similar modularity to the 500-L mAb facility, except it needs only two modules instead of three (Fig. 2B). The conventional platform's stainless-steel facilities are not modular and must be completely built before production begins. However, we assumed all these facilities can be used for multiple products even with single downstream trains through strategic campaigning. While the operational and logistical challenges to this modular, multi-reactor strategy are certainly not negligible and deserve further examination, for the purposes of this study we assumed that this approach can be implemented without significant difficulty.

2.3. Cost of goods

To understand business impact, we must first determine the costs of goods (COGs) for each facility type. Both operating expenses (OPEX) and capital expenses (CAPEX) are determined in this analysis. (OPEX consists of all ongoing expenses required to maintain a facility, such as costs for consumables, materials and labor. CAPEX

Table 2
Bioprocessing facility descriptions.

Platform	Product type	Bioreactor volume (L)	Upstream mode	Upstream material	Downstream mode
Continuous	mAb	500	Suspended perfusion	Single-use	Continuous
	non-mAb	500	Suspended perfusion	Single-use	Hybrid
Conventional	mAb	10,000	Fed-batch	Stainless steel	Batch
	non-mAb	2,000	Microcarrier perfusion	Stainless steel	Batch

Table 3
High-level process assumptions for both conventional and continuous platforms.

Parameter	mAb		non-mAb	
	10,000L stainless	500L continuous	2000L stainless	500L continuous
Avg. viable cell density (Mcell/mL)	12	120	5	60
Specific productivity (pg/cell/d)	35	35	10	10
Product titer (g/L)	5	2.1	0.05	0.6
Perfusion rate (RV/d)	–	2	1	1
Growth phase duration (d)	–	5	5	5
Production phase duration (d)	12	60	60	60
Reactor turnaround time (d)	2	1	2	1
Downstream capture	Batch	Continuous	Batch	Continuous
Downstream post-capture	Batch	Continuous	Batch	Batch
Product yield (%)	70	70	50	50

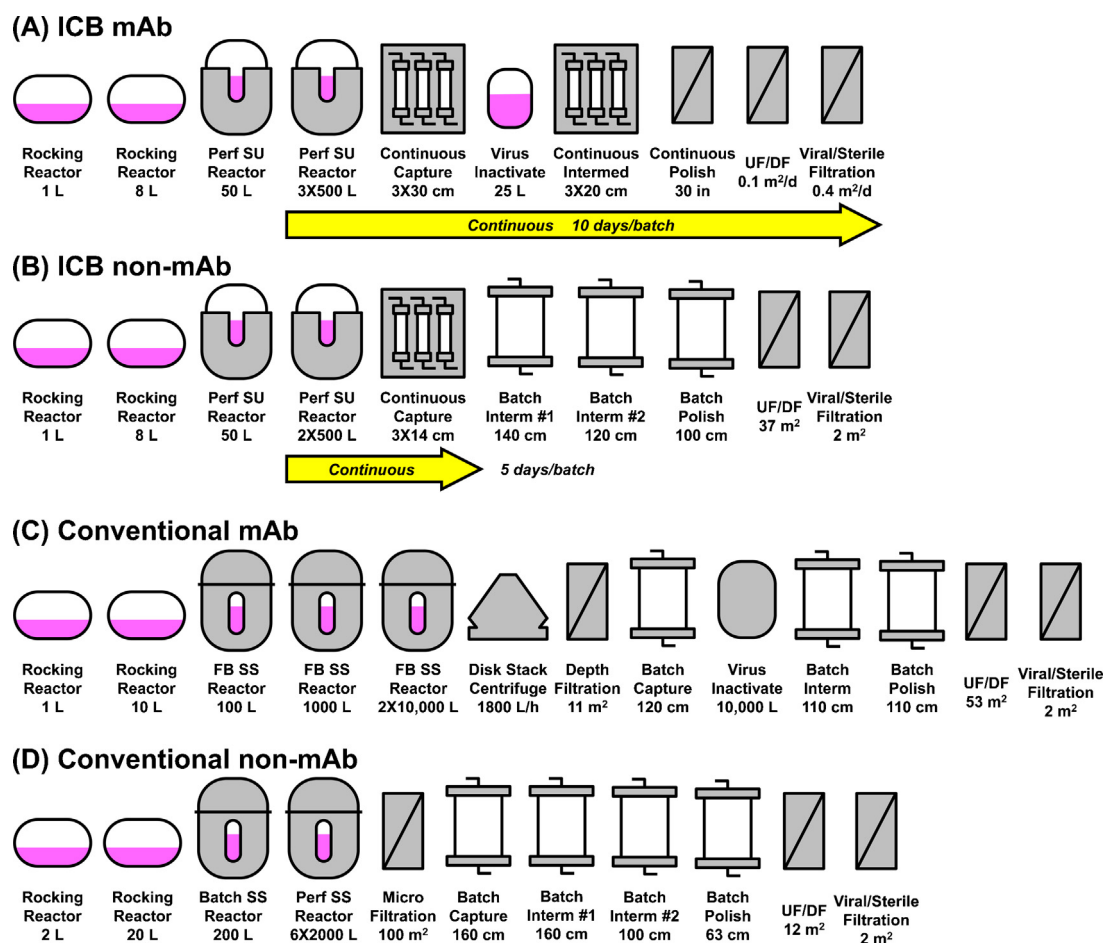


Fig. 1. Process definitions for the ICB and conventional platforms. (A) The ICB mAb process uses 500-L single-use production reactors coupled to cell retention and clarification technology. No stand-alone clarification unit operations are required. (We assumed that harvest yields are comparable across both ICB and conventional clarification.) Capture and intermediate purification steps are processed continuously, and multicycling membrane adsorbers carry out the polish step, followed by continuous filtration systems. (B) For the ICB non-mAb process, capture is also continuous, but capture eluate is pooled every five days and passed to a traditional batch downstream process with large columns and standard filtration. (C) The conventional mAb process consists of 10,000-L stainless-steel production reactors, devoted clarification unit operations and large batch columns. Large intermediate storage tanks are also required. (D) The conventional non-mAb process uses 2000-L stainless-steel perfusion bioreactors to meet demand. Harvest from the reactors are pooled for 10 days at a time, microfiltered and passed through an array of large column operations before various filtration steps. (Note that in this figure, chromatographic column sizes refer to diameters.)

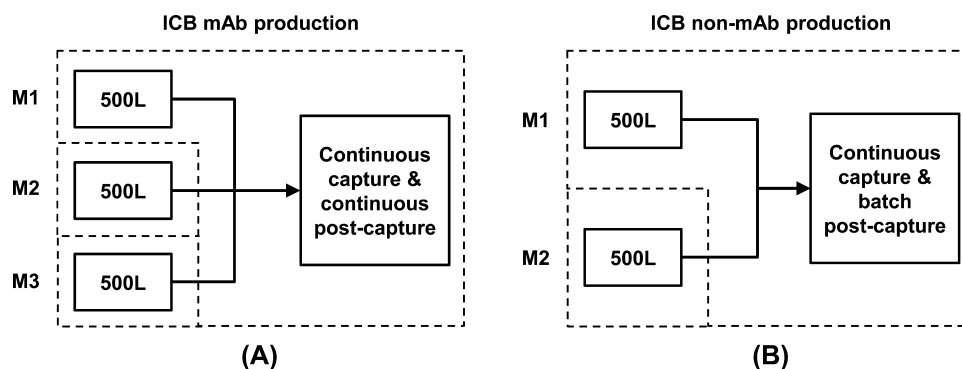


Fig. 2. Modularization approach for the single-use facilities. (A) For the ICB mAb facility, the first module consists of one reactor and the downstream train. Two additional modules can be built, each consisting of another 500-L reactor. (B) The continuous enzyme facility is similar, but each facility holds only two modules at a maximum.

refers to up-front costs to construct the facility.) We use the BioSolve cost modeling software (Biopharm Services, UK) to estimate COGs. Capital expenses around ancillary infrastructure, such as warehousing, research labs, pilot plants and office space were not considered here, since the types and costs of such support structures will vary widely across the industry according to different organizational philosophies. Indirect costs (such as validation costs) were also not included in this analysis. We also assumed no preexisting manufacturing capacity; new facilities are required to meet all product demand. We only considered operating expenses incurred directly from the manufacturing process; research and development, clinical trials and other costs are not included. The BioSolve process models have been updated to correctly reflect the ICB platform and unit operations, and OPEX and CAPEX calibrations were conducted to ensure that general cost assumptions were appropriate.

2.4. Risk-based NPV analysis

While COGs is an important indicator of business impact, to understand the overall picture, we also integrated the capital and operational expenses from COGs analysis with the product launch scenario to calculate time-based cash flows for each of the platforms. From this, we determined the NPV advantage (Δ NPV) of the ICB platform by calculating the differences in net present COGs between the novel platform and conventional platform. Fig. 3 presents an overview of the analytical architecture used to make these calculations.

NPV is a useful tool for decision making, particularly for capital investment and project approval and is popular for evaluating potential long-term projects because it accounts for up-front investment as well as the timing of future cash flows. Our NPV model considered costs of late-phase development, launch and stocking requirements, annual operating expenses and capital spending on the manufacturing network. To calculate operating expenses, we simply scaled the cost per unit product (as predicted by BioSolve) by the manufacturing throughput. We assumed that capital investment begins five years prior to launch for stainless steel facilities and 3.5 years for single-use facilities. Detailed investment profiles are shown in the Supplementary information (S2). We also assumed a weighted average cost of capital (WACC) of 7%, an inflation rate of 2% and a capital tax rate of 37%.

To evaluate the flexibility of the ICB platform, we also assessed the effect of risk and uncertainty on Δ NPV. In particular, we incorporated uncertainty around the following aspects:

- Technical transfer to manufacturing: There is always some risk when transferring processes from development to manufac-

turing, because scaling up can present unforeseen difficulties resulting in delays. The ICB platform has an advantage over conventional approaches here; because of its reduced volumes and footprint, no scale up is required between development and manufacturing. We incorporated the tech-transfer uncertainty using a simple probability of delay. For the conventional platform, we assumed a 50% chance that for each product there will be a tech transfer delay (6 months for mAb products and 12 months for non-mAb products). For the integrated continuous process, we assumed that the chance of a delay drops to 10% for each product. During technical transfer delays, additional development costs accrue and revenue is lost. (To model revenue loss, we price each drug at a conservative minimal break-even pricing such that total sales match total COGs.)

- Product failure during clinical trials: Usually, partial capital investment begins in Phase III stage of product lifecycle before a product is approved for commercial sales. Based on a recent study (Hay et al., 2014), we assigned mAb and non-mAb product success rates of 50 and 75%, respectively. On average, this will mean that three mAb and three non-mAb products are launched commercially over ten years. We assumed that manufacturing capacity will be built before product success is guaranteed; when products fail, facilities will simply sit empty until the next product successfully comes to market.
- Product demand profiles: Demand uncertainty is increasingly becoming a reality in the biotech industry. Before launch, each product was assumed to match a standard ramp of demand profile. Upon launch, we assigned a 15% probability that the peak demand could increase by 50% and a 15% probability that the peak demand could decrease by 20%. These profiles are described in the Supplementary information (S3).

We conducted a Monte Carlo experiment consisting of 10,000 discrete simulations using RiskAMP (Structured Data LLC, NY). We varied technical transfer delays, product failure and product demand per the probability distributions described above to generate risk-based Δ NPV for the ICB platforms (relative to conventional platforms). To analyze the drivers behind these results, we generated an overall Δ NPV distribution as well as sub-distributions by product type (mAb and non-mAb) and cost type (OPEX and CAPEX).

3. Results and discussion

3.1. Cost of goods

Within a given economic framework, process assumptions govern the output of a process-economic model. Using the previously mentioned process assumptions (Table 3) and economic assump-

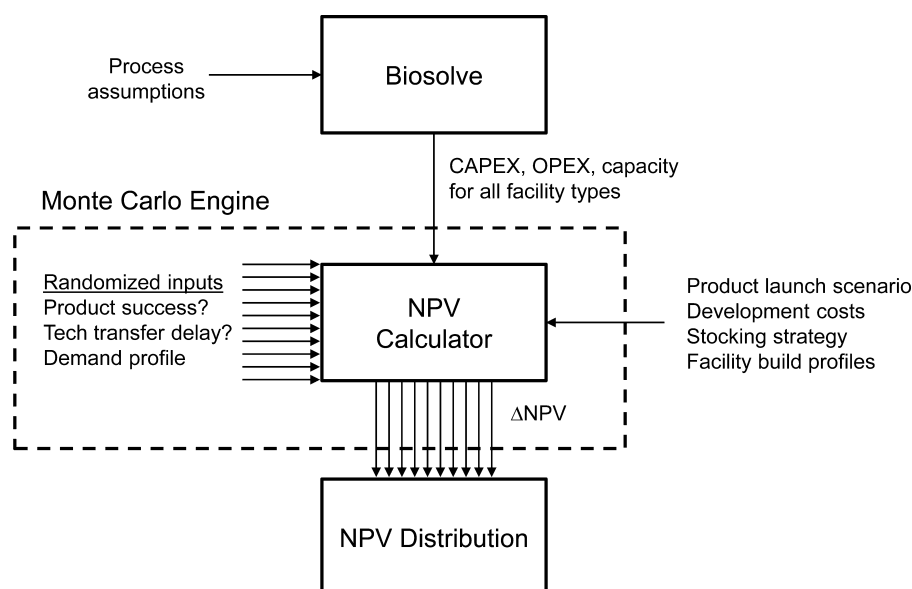


Fig. 3. Model architecture to estimate portfolio profitability. We fed process assumptions into BioSolve to model each bioprocess (ICB mAb and non-mAb, conventional mAb and non-mAb) and calculated CAPEX and OPEX for each corresponding facility. These values were fed into an NPV calculator, along with the product launch scenario and other business assumptions. To incorporate uncertainty, we ran Monte Carlo experiments to generate NPV distributions by conducting the NPV calculation thousands of times while varying product success, product demands and technical transfer success.

Table 4
OPEX and CAPEX estimations for each bioprocessing facility.

Platform	Facility	Reactors	Throughput (kg/y)	OPEX (\$/g)	CAPEX (\$M)
Continuous	mAb 500L	3 × 500 L	1236	17	58
Conventional	mAb 10kL	2 × 10,000 L	1537	22	110
Continuous	non-mAb 500L	2 × 500 L	84	242	63
Conventional	non-mAb 2kL	6 × 2,000 L	83	1232	229

tions within the BioSolve model, we estimated facility expenses (CAPEX and OPEX) for all processing platforms. Table 4 summarizes the results. OPEX and CAPEX estimations for mAb facilities are similar to those reported in recent literature (Kelley, 2009). Non-mAb facilities have significantly lower throughputs (approximately 200-fold) than mAb facilities due to lower demand. Because of lower productivities, non-mAb processes also have higher COGs (on a per-gram basis) by more than an order of magnitude. More importantly, comparing COGs across platforms for the same molecule reveals significant cost savings in moving toward the ICB platform. An ICB mAb facility reduces operating and capital expenses by 21 and 47%, respectively. An ICB non-mAb facility offers even more dramatic benefits, with OPEX and CAPEX reductions of 80 and 72%, respectively. These results are remarkable in light of the fact that conventional batch platforms have been developed extensively over last three decades and are considered mature in the industry. These results also agree with other models and real-world experiences suggesting that continuous processing offers significant benefits for biomanufacturing (Biopharm Services, 2014; EDB Singapore, 2014).

To further understand the key drivers behind the cost savings from the ICB platform, we divided OPEX and CAPEX into different upstream and downstream categories. (In this discussion, upstream operations refer to the seed train and production bioreactor, while downstream operations begin at the capture step and end at drug substance.) Results are shown in Figs. 4 and 5, respectively. ICB OPEX is lower for both mAb and non-mAb facilities compared to their corresponding conventional platforms. However, the breakdown of these savings is very different for the two classes of molecules. For mAb production in the ICB platform, increased costs for upstream filters and media are primarily due to higher media

consumption in a perfusion system relative to fed-batch. However, this increase is easily offset by several advantages in the ICB platform: (1) resin costs are significantly reduced due to the higher efficiency of continuous chromatography and the replacement of the polishing column with multicycle membrane adsorbers, (2) less upstream labor is necessary because there are fewer reactor turnarounds and (3) the continuous process does not require any standalone clarification operations. On the other hand, for non-mAb production, OPEX savings are present in every category, with the major drivers being (1) elimination of clarification operations, (2) reduction in downstream labor requirements, (3) reduced media consumption and (4) resin savings due to the efficiency of continuous capture chromatography.

Similarly, we also examined CAPEX across different unit operations (Fig. 5) for both mAb and non-mAb molecules. While OPEX savings are focused on only some unit operations, CAPEX savings can be found at every processing stage for both classes of molecules. Almost all unit operations contribute to the ICB platform's CAPEX advantage, with the major drivers being (1) large, expensive clarification equipment is not needed, (2) the continuous capture operation is smaller, (3) production bioreactors are smaller and disposable and (4) less frequent bioreactor turnaround leads to reduced seed train infrastructure. The ICB mAb process also replaces the polishing column with an intensified multicycle membrane adsorber integrated with intermediate purification, resulting in additional savings. Non-mAb facilities also reap additional CAPEX savings because they have higher productivities and therefore require fewer bioreactors than conventional facilities.

The OPEX and CAPEX savings described above provide detailed insights into benefits from a continuous bioprocess architecture; however, at a fundamental level, these benefits are integral to

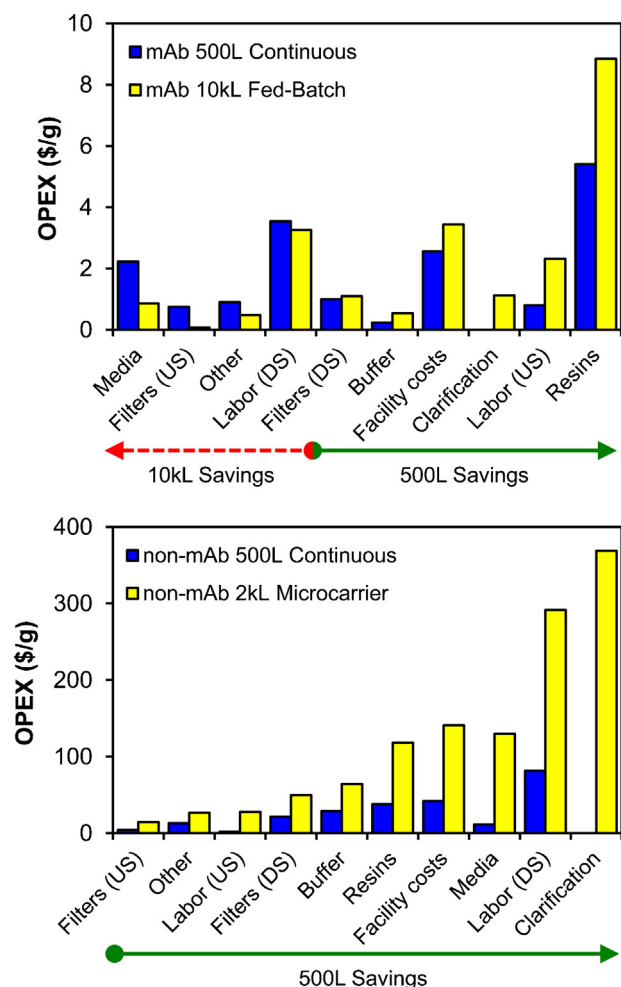


Fig. 4. OPEX breakdowns for the continuous and conventional platforms. Categories are arranged in order of increasing advantage to the continuous platform. This advantage is also indicated by solid (continuous advantage) and dashed (conventional advantage) arrows.

continuous processing in general. Several of these benefits arise from process intensifications in the ICB platform. The most important advantage comes from the platform's continuous nature—unit operations that are performed continuously can be greatly reduced in size. This temporal intensification most significantly benefits clarification in the continuous platform, shrinking this step so greatly that it can be merged completely into the production bioreactor operation. A similar shrinking occurs for intermediate hold tanks. Continuous intensification also benefits downstream purification steps. Columns larger than 100 cm in diameter can be replaced with multiple 30-cm columns (or smaller). These directly contribute to a reduction in facility footprint and therefore, lower CAPEX. Unit operation integration simplifies logistics and facility design. In theory, batch operations can achieve similar size reductions, but in so doing would trigger significantly higher labor costs. The ICB systems are designed to operate continuously with appropriate automation and control, achieving this intensification without corresponding labor increases.

However, the continuous platform also realizes other, less intuitive forms of process intensification that are dependent on specific unit operations. The technical nature of these intensifications typically prevents their use in batch processing. For example, continuous upstream bioreactors can be run at 10–15-fold higher cell densities than batch processing, reducing reactor number and/or volume. Continuous chromatography can also more efficiently uti-

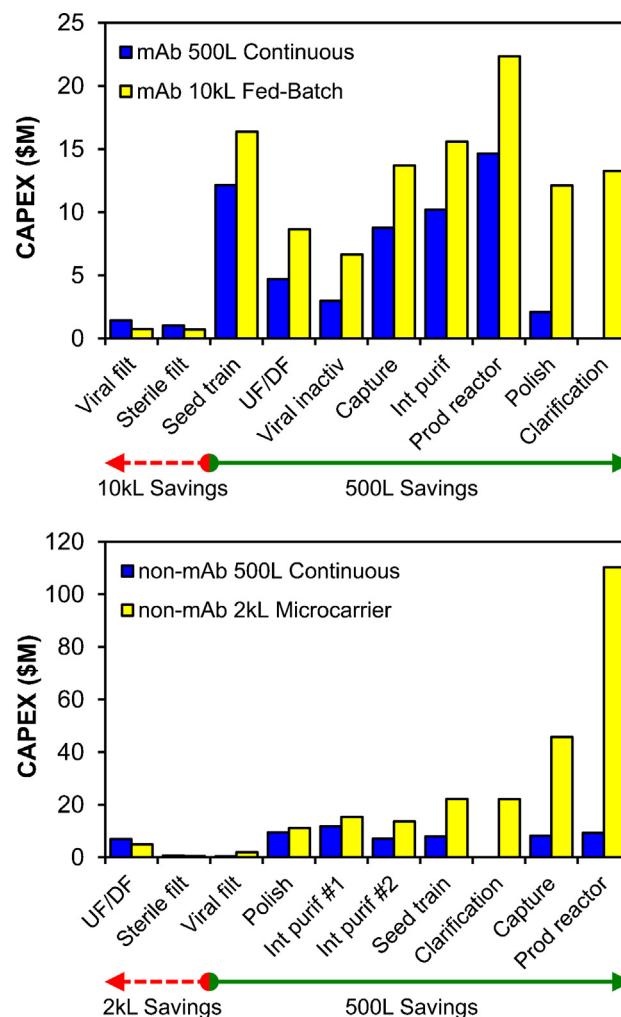


Fig. 5. CAPEX breakdowns for the continuous and conventional platforms. Categories are arranged in order of increasing advantage to the continuous platform. This advantage is also indicated by solid (continuous advantage) and dashed (conventional advantage) arrows.

lize expensive resins. All together, these process intensifications are responsible for the cost gains achieved by the ICB platform.

3.2. Platform NPV analysis

Cost of goods analysis provides a good picture at a single product level; however, the goal of any biotech company is to manufacture and supply a portfolio of products over a long period of time. Therefore, selection of bioprocessing platform technologies needs to be driven by the product portfolio. For quantitative decision making, the COGs analysis shown above for the ICB platform needs to be converted into cash flows and net present value for its entire product portfolio.

NPV analysis allows operating and capital expenses to be combined along with uncertainty to enable a holistic evaluation of biomanufacturing platforms. NPV analysis can also take into account modular facilities that are built as needed over time. For the single-use ICB processes, facilities can evolve over time as modules are added. To more accurately assess NPV, we estimated OPEX and CAPEX for each stage of modular construction and included this in the overall analysis. Cost details for each modularization level are shown in the Supplementary information (S4).

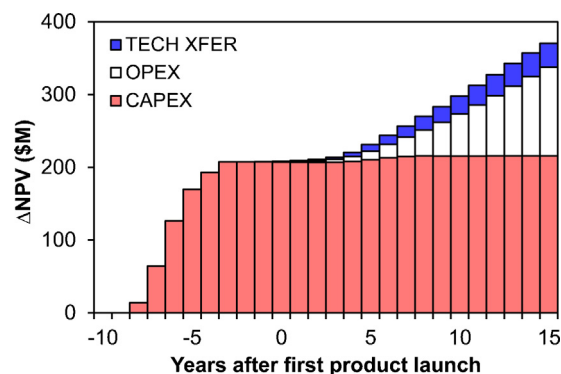
For the hypothetical product portfolio selected in this work, the continuous platform presents a significant and compelling advan-

Table 5 Δ NPV comparison for different facilities (relative to the conventional platform).

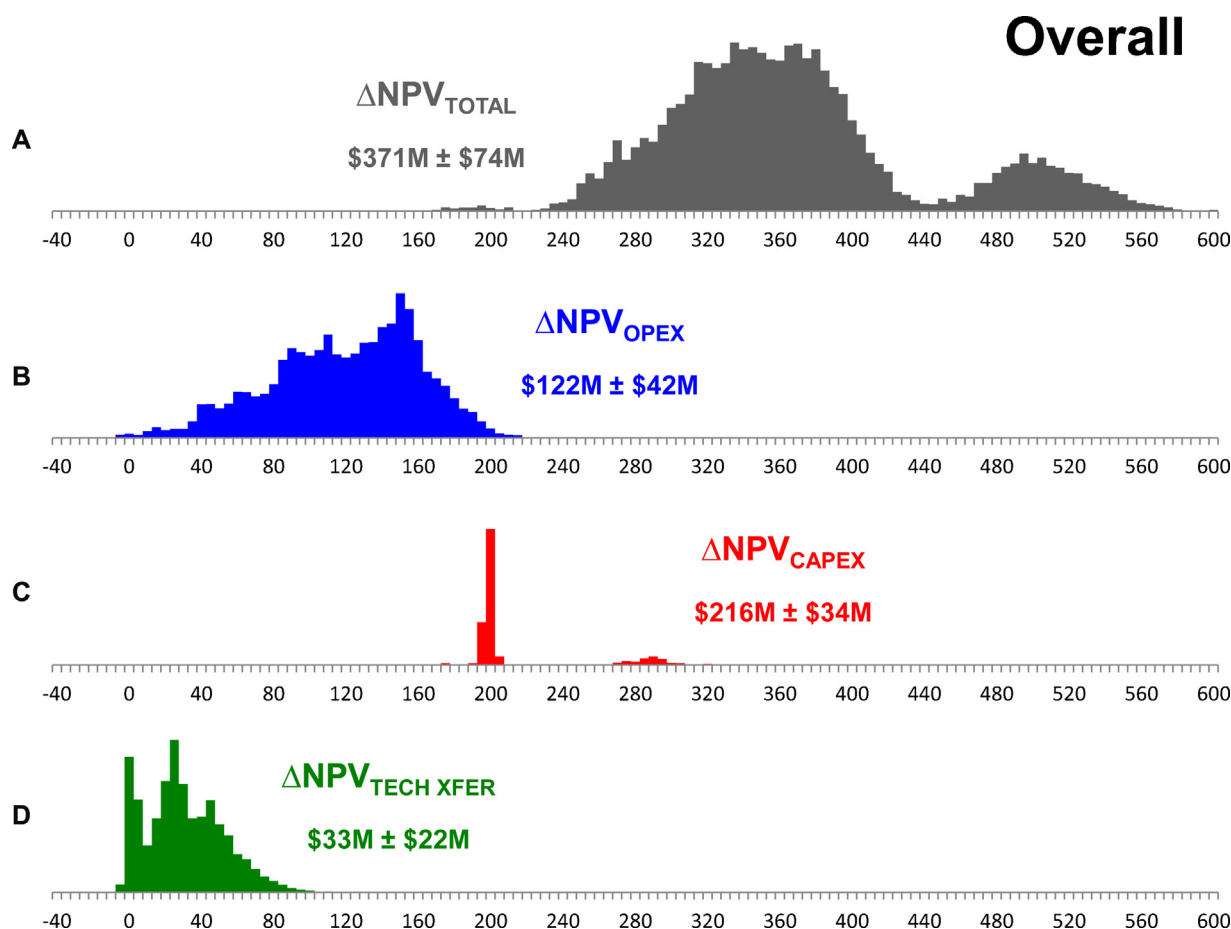
Savings from:	Δ NPV OPEX (\$M)	Δ NPV CAPEX (\$M)	Δ NPV TECH XFER (\$M)	Δ NPV Total (\$M)
Continuous production of mAbs & non-mAbs	122	216	33	371
Continuous production of mAbs	3	56	6	64
Continuous production of non-mAbs	119	160	27	306

tage over conventional bioprocessing. Table 5 summarizes these results. On average, the CAPEX and OPEX savings of the ICB platform represent a 55% reduction in cost relative to the conventional platform. Together, the CAPEX, OPEX and technical transfer advantages of the continuous platform translate to an NPV advantage (Δ NPV) of \$371 M. To put this number in perspective, these savings are sufficient to construct six additional continuous facilities (supplying an additional 7000 kg/y of mAb or 500 kg/y of non-mAb). A chart of cash flows in Fig. 6 demonstrates how those savings are realized over time. The ICB platform almost immediately begins accruing benefits (cash flows become positive) due to the earlier and relatively more costly stainless-steel construction required by the conventional platform. These CAPEX savings begin to plateau at about \$200 M of present value in five years, after which additional OPEX and technical transfer contributions accumulate fairly linearly.

Besides the average output, Monte Carlo simulations also convert uncertainty into output distributions that can aid strategy and decision making. The Δ NPV distributions are displayed in Fig. 7 for the ICB platform. The overall Δ NPV distribution (Fig. 7A) has a fairly wide range, with 99% of all cases ranging between \$230 M

**Fig. 6.** Δ NPV over time, divided into CAPEX, OPEX and technical transfer contributions.

and \$600 M. This result shows that in virtually all scenarios, the ICB platform retains its significant financial advantage over the conventional platform. Interestingly, the Δ NPV distribution is bimodal, with broad peaks at \$350 M and \$500 M. For further understand-

**Fig. 7.** Monte Carlo distributions for NPV differences between the continuous platform and the conventional platform. Distribution means and standard deviations are also noted.

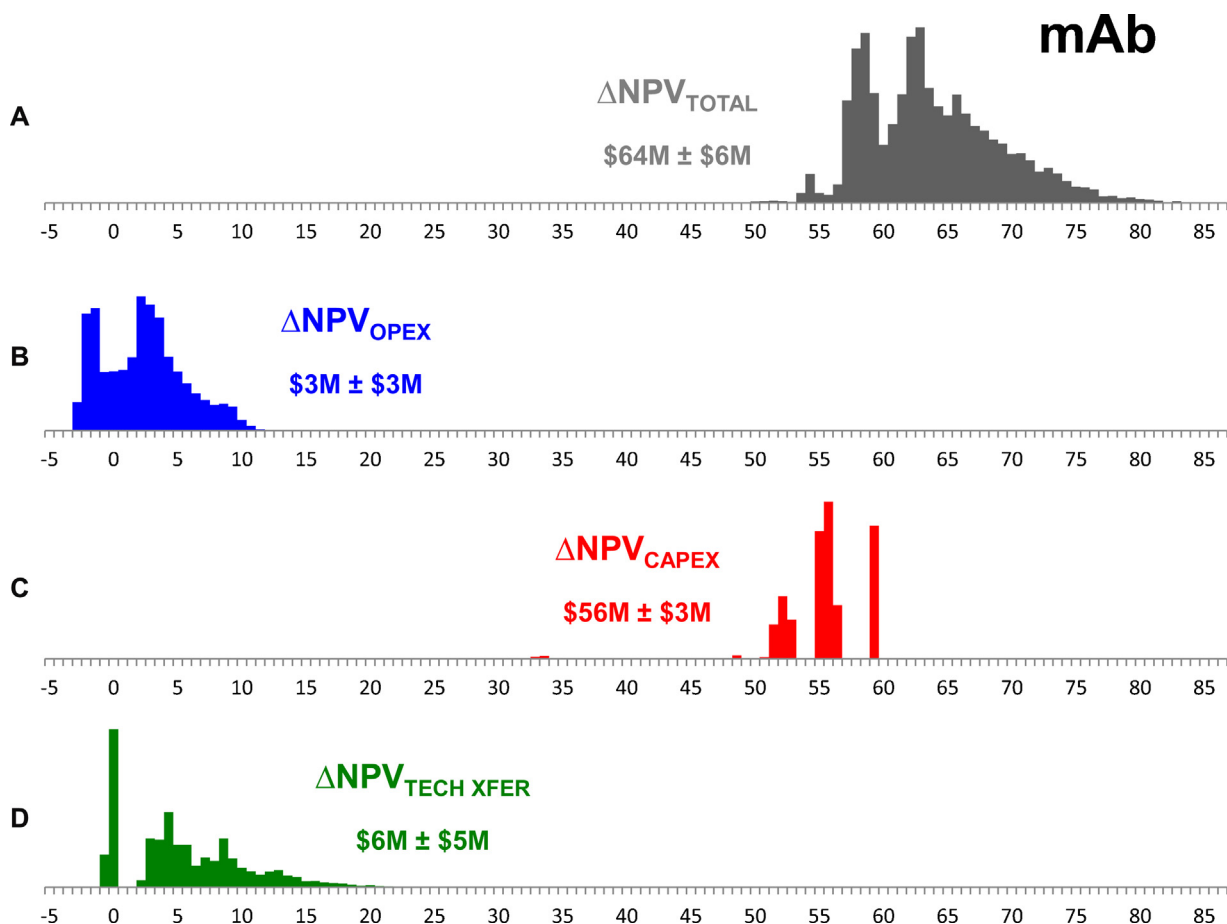


Fig. 8. Monte Carlo distributions for NPV differences between the continuous mAb facility and the stainless-steel fed-batch mAb facility. Distribution means and standard deviations are also noted.

ing, we broke down different contributions to this overall ΔNPV (Fig. 7B–D). OPEX and tech transfer distributions are fairly smooth showing that 10,000 simulation data points are sufficient to investigate the model variability. Since CAPEX events are discrete, a smooth CAPEX distribution cannot be expected. The bimodal nature of the overall ΔNPV is mostly driven by the CAPEX contribution with sharp peaks around \$200 M and \$300 M.

To better understand the driving forces behind these distributions, we can consider mAb and non-mAb production in each cost category. Fig. 8 shows different ΔNPV distributions for mAb production. The majority of savings in the ICB mAb platform results from reduced CAPEX due to reduced equipment needs (in number and in size). The CAPEX breakdown (Fig. 8B) has three distinct peaks, consistent with the three-step modularization strategy used by the ICB platform, where 500-L bioreactors are added to the facility as product demand increases. The contribution coming from tech transfer and OPEX are comparatively smaller (Fig. 8A and C). Further analysis of OPEX contribution over every simulation point reveals that OPEX contribution to cost savings typically increases with increased mAb demand (Fig. 9). This is not surprising: as more product is made, more savings accrue due to the fact that the ICB platform has lower operating expenses than the conventional platform. Tech transfer savings accounted for approximately 9% of total savings, and this contribution could potentially be even higher when realistic revenue and competition models are considered for these types of biotech products. However, because we chose to focus only on biomanufacturing-driven benefits, for this study we assumed a simplistic revenue model without considering pricing strategies and competition.

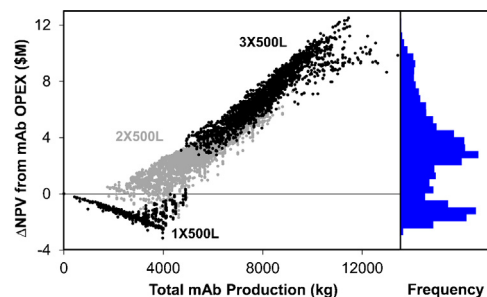


Fig. 9. OPEX savings from continuous mAb production versus total mAb production over the product portfolio. A histogram also links the mAb OPEX savings back to the ΔNPV distribution in Fig. 7. mAb OPEX savings primarily correlate to the total amount of mAb produced over the 25-year scenario, but there is also a sub-correlation between OPEX savings and the final modularization of the ICB platform. As facility utilization increases (towards the maximum capacity of three 500-L bioreactors), it operates at its highest efficiency and savings increase most steeply with production. The three modularization levels (1 $\times$ 500 L, 2 $\times$ 500 L, 3 $\times$ 500 L) are indicated both with text and shade.

Similar breakdowns for non-mAb products (Fig. 10) show that ICB platform savings result from both from OPEX and CAPEX advantages with comparatively smaller contributions from tech transfer advantages. The CAPEX contribution for non-mAb is bimodal and larger than its mAb counterpart; therefore, it drives the bimodal nature of the overall ΔNPV . This occurs because in the higher non-mAb demand scenarios, additional facilities must be constructed, resulting in a second, higher ΔNPV peak. As with mAb production, OPEX savings correlate closely to cumulative production (Fig. 11).

non-mAb

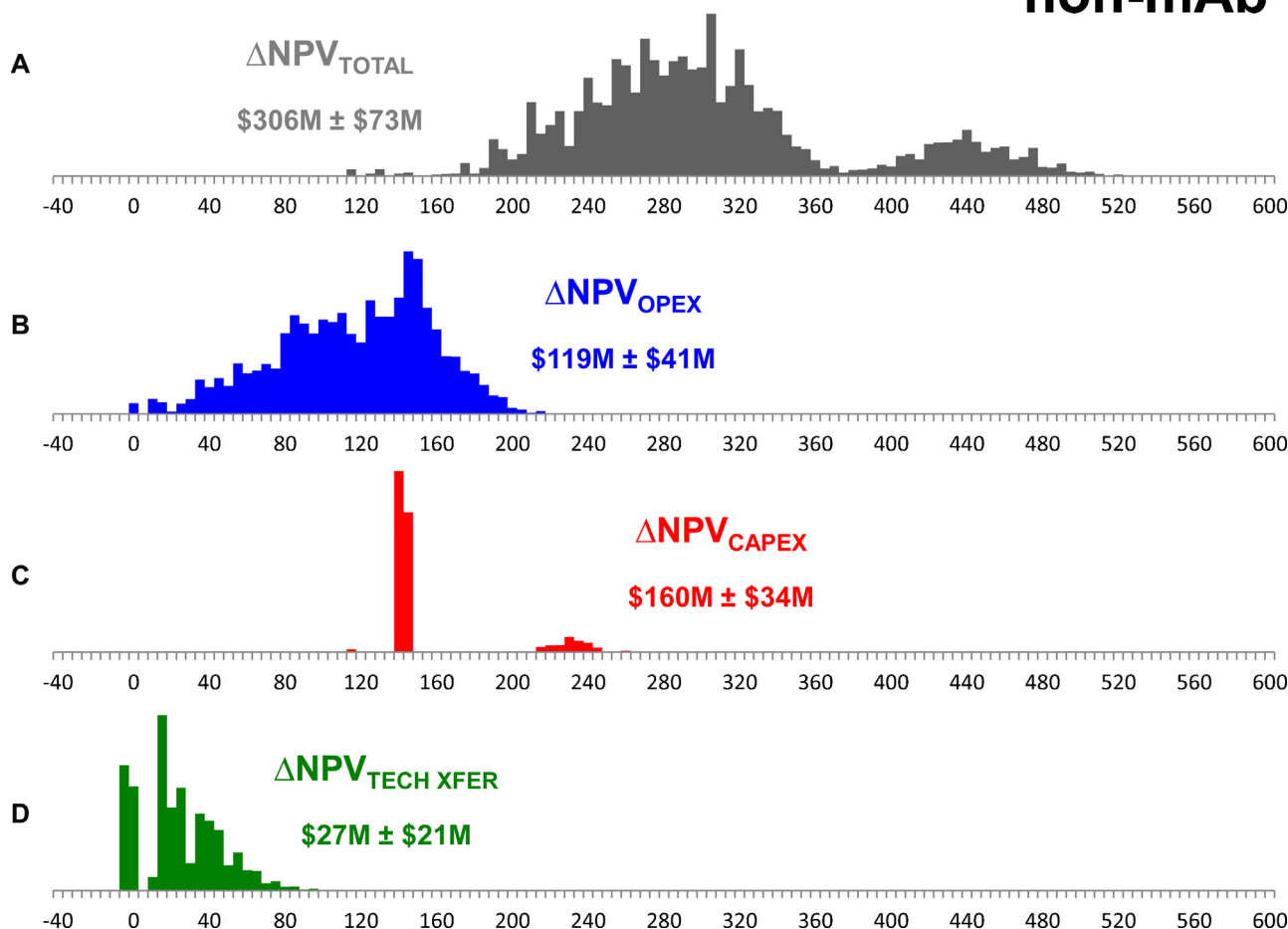


Fig. 10. Monte Carlo distributions for NPV differences between the continuous non-mAb facility and the stainless-steel microcarrier non-mAb facility. Distribution means and standard deviations are also noted.

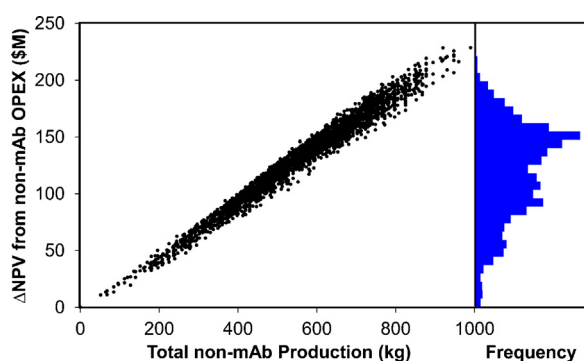


Fig. 11. OPEX savings from continuous non-mAb production versus total non-mAb production over the product portfolio. A histogram also links the mAb OPEX savings back to the Δ NPV distribution in Fig. 9. Non-mAb OPEX savings primarily correlate to the total amount of non-mAb produced over the 25-year scenario. Modularization level is less important than with mAb production, because the continuous platform has significantly lower operating expenses even when facilities are under capacity.

3.3. Single-use fed-batch platform

Single-use fed-batch facilities offer simplified and accelerated product changeover, and are well suited for clinical manufacturing, due to the high rate of product attrition in clinics (Eibl et al., 2010; Shukla and Gottschalk, 2013). With increasing cell culture titers, single-use fed-batch facilities are also becoming

attractive for small to medium-scale commercial mAb production. Previous work has shown that there is typically a batch number threshold below which these facilities are cost-effective for commercial production, because increased consumable costs are offset by greater reductions in other areas, such as energy, water and labor (Sinclair and Lim, 2007). Beyond a certain batch number, these facilities lose their cost advantage due to the high cost of consumables. Could a single-use fed-batch platform cost less than the ICB platform? Recent studies have shown that continuous facilities can offer more financial advantages than single-use facilities for mAb production (Biopharm Services, 2014). We also find similar results when comparing the ICB mAb platform with a 2000-L single-use fed-batch mAb platform: the ICB platform still offers 57% CAPEX and OPEX savings (across the entire product portfolio) relative to a batch platform relying on single-use fed-batch reactors for mAb production and stainless steel microcarrier reactors for non-mAb production. In absolute terms, the ICB platform delivers \$335 M in savings. Just considering mAb production, the ICB platform offers 18% combined CAPEX and OPEX savings (\$29 M). We believe the ICB platform uniquely combines the benefits of single-use systems (lower CAPEX, easy product changeover) and stainless steel (lower consumable costs due to infrequent changeover) and therefore, these results can be intuitively expected. Additional information about the comparison with 2000-L fed-batch platform can be found in the Supplementary information (S5).

4. Conclusions

The current market drivers in the biotech industry require process innovation to increase manufacturing flexibility and decrease COGs. In this manuscript, we have shown that continuous bioprocessing not only meets these current business needs but can position the biotech industry to expand its promise to serve even larger populations and unmet medical needs. We introduced an integrated continuous biomanufacturing platform and directly compared its financial performance to conventional biomanufacturing technologies. We conducted this analysis using process-economic modeling tools, Monte Carlo simulations and risk-based net present value analysis. We developed a novel mathematical framework to understand the financial impact of moving toward continuous bioprocessing. Our work combines process-economic models of multiple types of products, a complex product portfolio over time and various business and technical uncertainties to yield a comprehensive risk-based NPV output. This output can reveal drivers and aid decision making for selection and development of continuous bioprocessing technology. The ICB platform offers significant financial advantages due to multiple process intensification (such as smaller, fewer and more efficient unit operations, more flexible facilities, reduced turnaround time and increased automation) leading to hundreds of millions of dollars in savings.

In addition to the quantitative advantages we demonstrated here, the continuous platform offers even more upward potential. There are other intangible benefits, including steady-state operation, reduced cycle times, and steady-state product quality which were not quantified here and can be investigated in the future. Another example of an intangible benefit is the potential universality of the ICB platform. While two somewhat different continuous approaches (fully continuous and hybrid continuous) were presented here for mAb and non-mAb manufacturing, these two process architectures could converge in the future and be consolidated within the same facility, offering even greater flexibility and savings.

Appendix A. Supplementary information

Supplementary information associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jbiotec.2015.05.010>

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